

Recording

Date	:	17/06/2026
Problem Specification	:	Develop a Java program that allows the user to enter a set of integers, search for a specific number and then sort the array in ascending order.
Assumption	:	1. Assume the user enters a valid positive integer for the array size. 2. The array contains at least one element.
Limitation	:	1. The program only accepts integer values. 2. If the same number appears multiple times, only the first occurrence is reported.
Input	:	1. Number of elements in the array. 2. Array elements. 3. Target value to search.
Processing	:	Take array from the user as the input, then perform linear search to find the number and return It's index, then sort the array in ascending order using bubble sort.
Output	:	1. Original array. 2. Search result (found/not found and index position). 3. Sorted array in ascending order.
Algorithm	:	Step1: Read the number of elements (num) Step2: Create an integer array (arr[]) Step3: Read (num) elements into (arr[]) and store all array elements. Step4: Display the original array. Step5: Read the target value to search. Step6: Set $i \leftarrow 0$. Step7: Compare arr[i] with target. Step8: If arr[i] = target, return i as the index. Step9: Increment $i \leftarrow i + 1$. Step10: If $i < \text{arr.length}$, go to Step 7. Step11: Return -1 (target not found). Step12: If returned index $\neq -1$, display "Found target at index " Step13: Otherwise, display "target not found". Step14: Set outer counter $i \leftarrow 0$.

Step15: Set inner counter $j \leftarrow 0$.
Step16: If $\text{arr}[j] > \text{arr}[j+1]$, swap the two elements.
Step17: Increment $j \leftarrow j + 1$.
Step18: If $j < \text{arr.length} - i - 1$, go to Step 16.
Step19: Increment $i \leftarrow i + 1$.
Step20: If $i < \text{arr.length} - 1$, go to Step 15
Step21: Display the sorted array.
Step22: End program.

Program listing : Program file attached

Test data and expected output : 1. Test data: num = 5
Enter 5 numbers:
8
3
9
7
1
Enter Number to search: 9

Expected output: Original Array: 8, 3, 9, 7, 1
Found 9 at index 2

Sorted Array: 1, 3, 7, 8, 9

2. Test data: num = 4
Enter 4 numbers:
5
9
7
2
Enter Number to search: 11

Expected output: Original Array: 5, 9, 7, 2
11 not found!

Sorted Array: 2, 5, 7, 9

Output obtained for test data :

1. Test data: num = 5
 Enter 5 numbers:
 8
 3
 9
 7
 Enter Number to search: 9

Obtained output: Original Array: 8, 3, 9, 7, 1
 Found 9 at index 2

Sorted Array: 1, 3, 7, 8, 9

2. Test data: num = 4
 Enter 4 numbers:
 5
 9
 7
 2
 Enter Number to search: 11

Obtained output: Original Array: 5, 9, 7, 2
 11 not found!

Sorted Array: 2, 5, 7, 9

Analysis : The numbers of operation required in performing the algorithm.

	+, -	/, *	%	</>/<=>=
Linear Search	-	-	-	$\leq N$ times
Bubble Sort	-	-	-	$N(N-1)/2$ times

Conclusion : This program allows users to search for a specific number in an array and sort the array in ascending order. The implementation helps in understanding Linear Search and Bubble Sort techniques in Java.

Discussion : The linear search has worst-case time complexity of $\Theta(N)$, the bubble sort has worst-case time complexity of $\Theta(N^2)$ because of its nested loops. For better efficiency we can use better sorting algorithms like Merge sort.